MIVA OPEN UNIVERSITY

Submitted By:

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INDIVIDUAL PRACTICAL DATABASE PORTFOLIO

Selected case study: SwiftBank

GITHUB REPOSITORY

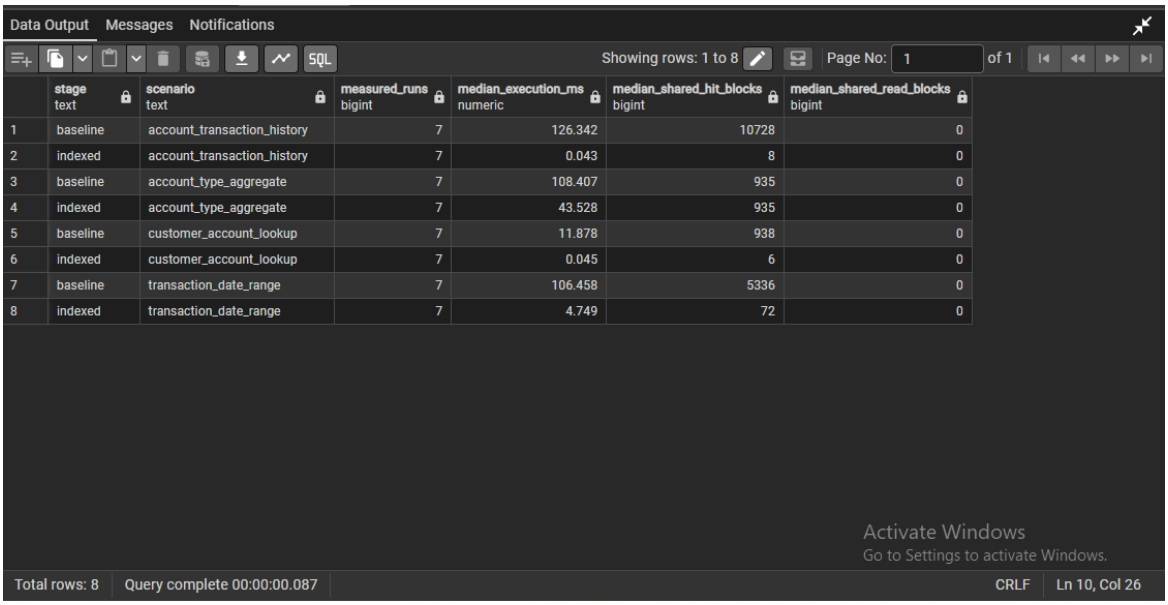
https://github.com/femenike0518-wq/advanced-database-portfolio

Selected case study and portfolio results

My made-up banking system for customers with current and savings accounts is called SwiftBank. The database is used by administrators, bank employees and customers. Customer contact information, accounts, balances, transaction types, amounts, statuses, dates and account references are all stored there. Account opening and maintenance, deposits, withdrawals, transfers, transaction history, viewing and transaction review are all included in the supported work.

Portfolio	Work and Recorded Result
P1 (Design)	For customers, accounts and transactions, I created a relational model. Six customers, six accounts and eleven transactions make up my repeatable seed; the verification records balance that match completed activity.
P2 (Queries)	I benchmarked 500,000 transactions, 100,000 accounts and 50,000 customers. Customer lookup 11.878 to 0.045, account-type aggregate 108.407 to 43.528, history 126.342 to 0.043, and date report 106.458 to 4.749 are the baseline to indexed medians (ms). The aggregate remained a sequential scan because all four result signatures matched.
P3 (Concurrency)	Instead of the anticipated 4,000.00 from a starting balance of 4,250.00, two stale withdrawals left 4,150.00. Both withdrawals were applied in the separate guarded run and the final balance was 3,900.00.
P4 (No SQL)	Three SwiftBank customers with embedded accounts, Decimal128 balances and bounded seed transaction histories were modeled in MongoDB. PostgreSQL is still in charge of balances and money transfers, so I kept the unsuccessful withdrawal with a failed status.
P5 (Distributed Cloud)	I've created a different async standby for PostgreSQL 18.6 locally. One standby write was rejected, the marker row was replicated and the primary reported streaming. Failover, cloud options and sharding were only discussed in the work that was recorded.

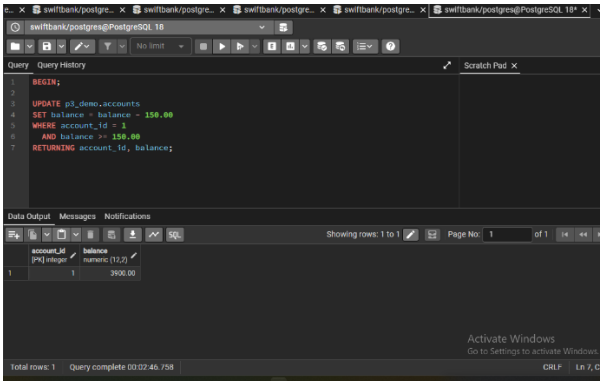
Screenshot



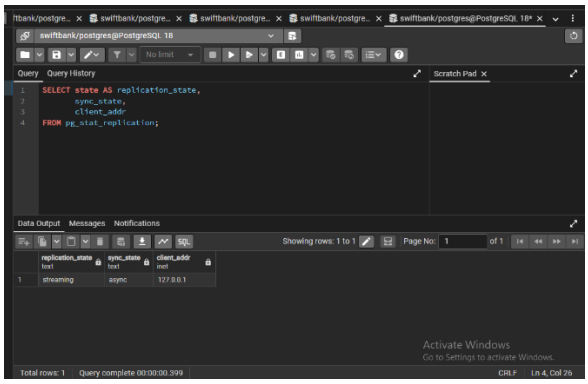
P2 median summary: Each stage involved seven measured runs and the screenshot was replicated using repository evidence.

Technical Reflection

I can share some of the PgAdmin queries I used and the outcomes they gave me. I was still getting used to things and had mixed up the scripts, so at first I didn't get the results I wanted. After receiving some assistance, I discovered how to approach the portfolio in a step-by-step manner. The number of useful database tasks that PowerShell and CMD could perform was unexpected. The lost update result is the best example of what I discovered. Instead of the 4,000.00 you would anticipate if both withdrawals were applied, two stale writes left 4,150.00. If both withdrawals were applied to the initial 4,150.00 balance in a different guarded run, you would ultimately get 3,900.00.



P3: the guarded withdrawal run ends at 3,900.00



P5: the primary returns one streaming asynchronous standby

AI-use declaration

I used ChatGPT to fully understand the assignment and the work I needed to complete. I used Codex to create and evaluate my portfolios, correct any mistakes and complete my final documentation.

References

- The PostgreSQL Global Development Group. EXPLAIN is used in PostgreSQL 18 documentation. https://www.postgresql.org/docs/18/using-explain.html
- The PostgreSQL Global Development Group. Log-shipping standby servers in PostgreSQL 18 documentation. The URL is https://www.postgresql.org/docs/18/warm-standby.html
- MongoDB. Types of BSON: Decimal128. https://www.mongodb.com/docs/manual/reference/bson-types/
- Schema validation in MongoDB. https://www.mongodb.com/docs/manual/core/schema-validation/